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Article

Boosting MS1-only Proteomics with Machine Learning Allows 2000 Protein Identifications in Single-Shot Human Proteome Analysis Using 5 min HPLC Gradient

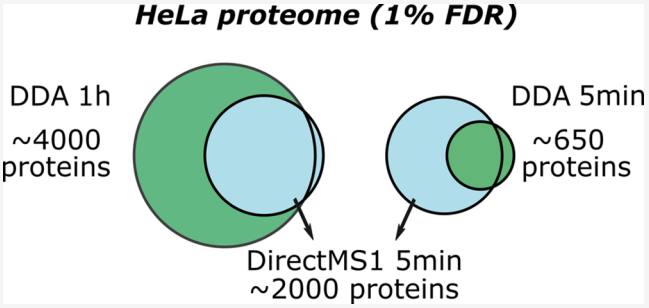
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ABSTRACT: Proteome-wide analyses rely on tandem mass spectrometry and the extensive separation of proteolytic mixtures. This imposes considerable instrumental time consumption, which is one of the main obstacles in the broader acceptance of proteomics in biomedical and clinical research. Recently, we presented a fast proteomic method termed DirectMS1 based on ultrashort LC gradients as well as MS1-only mass spectra acquisition and data processing. The method allows significant reduction of the proteome-wide analysis time to a few minutes at the depth of quantitative proteome coverage of 1000 proteins at 1% false discovery rate (FDR). In this work, to further increase the capabilities of the DirectMS1 method, we explored the opportunities presented by the recent progress in the machine-learning area and applied the LightGBM decision tree boosting algorithm to the scoring of peptide feature matches when processing MS1 spectra. Furthermore, we integrated the peptide feature identification algorithm of DirectMS1 with the recently introduced peptide retention time prediction utility, DeepLC. Additional approaches to improve the performance of the DirectMS1 method are discussed and demonstrated, such as using FAIMS for gas-phase ion separation. As a result of all improvements to DirectMS1, we succeeded in identifying more than 2000 proteins at 1% FDR from the HeLa cell line in a 5 min gradient LC-FAIMS/MS1 analysis. The data sets generated and analyzed during the current study have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository with the data set identifier PXD023977.



KEYWORDS: *protein identification, mass spectrometry, high-throughput proteomics, machine learning, retention time prediction*

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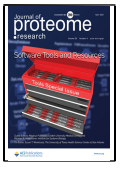
INTRODUCTION

High-throughput and sensitive analytical approaches enabling proteome-wide measurements of large cohorts of samples will open the way for the application of mass-spectrometry-based proteomics in clinical trials, population proteomics, and the emerging areas of drug-to-proteome interactions and meta-proteome characterization.[1,2] Moreover, the needed throughput and depth of proteome coverage in these studies has to be accompanied by protein quantitation that is consistent across the sample cohorts, which is necessary for these approaches to be useful in personalized medicine studies.[3,4] A recent example of using the large cohort of COVID-19 patients clearly demonstrated the need for ultra-high-throughput proteomics to generate hypotheses about therapeutic targets and to aid classification or diagnostic decision making in clinical environments.[5]

The typical LC-MS/MS-based proteomic methods are extremely time-consuming, which is partially overcome by expensive multiplexing using isotopic labeling strategies, such as tandem mass tag (TMT).[6] Nevertheless, a number of recent studies employing the bottom-up proteomic approaches have become more focused on increasing the throughput of the proteome-wide analysis. These studies include all steps of the proteome characterization workflow, such as increasing the speed of sample preparation prior to MS-based analysis,[7−11] squeezing the data acquisition time to the few minutes range,[12−14] and increasing the throughput of data processing.[15−20] The most recent and extreme examples include the proposal for LC-free MS/MS-based proteome analysis using direct infusion of the proteolytic mixture, which demonstrated the identification of 500 proteins within minutes of data acquisition.[21]

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One of the most time-consuming parts of the analysis is MS/MS data acquisition, which requires long HPLC gradient times to have enough room for the sequential isolation of as many as possible precursor ions from each MS1 spectrum for subsequent fragmentation. A number of methods and approaches to MS/MS-free proteome analysis have been proposed and widely explored starting from the early Accurate Mass and Tag (Time) method[22−26] to a recent truly (e.g., without employing tandem mass spectrometry at any step in the workflow) MS/MS-free realization.[27,28] These methods rely heavily on the accuracies of both peptide m/z measurements and retention time (RT) predictions. The latter is especially important for the MS/MS-free strategy, as RTs contain the only sequence-specific information in a (m/z, RT) space.[29,30] With advances in the development of machine-learning algorithms, a variety of highly accurate RT prediction models become increasingly available.[31−34] The work of Bouwmeester et al.[34] is particularly interesting because it shows that DeepLC model's performance is comparable to that of the other deep-learning-based alternatives, yet it provides better generalization between different chromatography setups. This feature is particularly useful for MS/MS-free approaches such as DirectMS1 when the number of peptides available for RT prediction model training is typically small.

Previously, we described the DirectMS1 method, in which proteolytic peptide mixture analysis was performed in the MS/MS-free mode of acquisition using high-resolution mass spectrometry.[27] Because the method does not employ isolation and fragmentation steps, the time for LC peptide separation can be significantly reduced to a few minutes, and the number of MS1 spectra available for processing will be limited only by the acquisition rate of the mass analyzer operating at high-mass-resolution settings. For the first time, DirectMS1 demonstrated the capability to identify up to 1000 proteins from a mammalian cell sample using only a 5 min HPLC gradient. Moreover, the average sequence coverage for each identified protein in this method significantly exceeded the one of a standard MS/MS-based approach, which further contributed to improved quantitation. On a proteome-wide scale, this kind of analysis efficiency was not considered feasible before because of the whole proteome sample complexity, a lack of sequence-specific information in the measured m/z values of peptide ions, and the low accuracy of the previous RT prediction models. However, advances in high-resolution mass spectrometry technology and machine-learning-based RT prediction models implemented in DirectMS1's search engine allowed the method to exceed the proteome coverage rate per analysis time of long-gradient MS/MS-based proteomics. Note, however, that the previously achieved DirectMS1 method's efficiency did not reflect the real information content of the MS1 data set, in which all eluted and ionizable peptides of the analyzed proteolytic mixture are potentially present in a full dynamic range accessible by current state-of-the-art high-resolution mass analyzers, leaving further room for improvements.

In this work, we integrated two novel machine-learning algorithms into the data processing workflow of DirectMS1 to further enhance its efficiency. These algorithms include the previously mentioned DeepLC as an RT prediction model[34] and the Light Gradient Boosting Machine (LightGBM)[35] used for scoring peptide feature matches. Also, the method was upgraded and tested for processing MS1-only data obtained using high-resolution mass spectrometry combined with high-field asymmetric waveform ion mobility (FAIMS),[36] which has become increasingly popular in proteomic research[37,38] because it provides an additional separation dimension for peptides at the front end of a mass spectrometer.

EXPERIMENTAL SECTION

Samples

All experiments were performed using Thermo Scientific Pierce HeLa protein digest standard (P/N 88328) derived from the HeLa S3 cell line.

Data Acquisition

LC-MS analysis in this work was performed using an Orbitrap Fusion Lumos mass spectrometer (Thermo Scientific, San Jose, CA) coupled to an UltiMate 3000 LC system (Thermo Fisher Scientific, Germering, Germany) and equipped with the FAIMS Pro interface. A short LC gradient method was applied, as previously reported.[27] A μ-Precolumn C18 PepMap100 trap column (5 μm, 300 μm i.d., 5 mm, 100 Å) (Thermo Fisher Scientific, USA) and a self-packed analytical column (Inertsil 3 μm, 75 μm i.d., 15 cm length) were employed for separation. The mobile phases were as follows: (A) 0.1% formic acid (FA) in water and (B) 80% acetonitrile (ACN) and 0.1% FA in water. The loading solvent was 0.05% trifluoroacetic acid (TFA) in water. The gradient was from 5 to 35% phase B in 4.8 min at 1.5 μL/min. The total method time including the column washing and equilibration was 7.3 min. Field asymmetric ion mobility spectrometry (FAIMS) separations were performed with the following settings: inner and outer electrode temperatures were 100 °C, FAIMS carrier gas flow was 4.7 L/min, asymmetric waveform dispersion voltage (DV) was −5000 V, and entrance plate voltage was 250 V. Compensation voltages (CVs) of −50, −65, and −80 V were used in a stepwise mode during LC-MS analysis. Mass spectrometry measurements were performed in the MS1-only mode of acquisition. Full MS scans were acquired in a range from m/z 375 to 1500 at a resolution of 120 000 at m/z 200 with an automatic gain control (AGC) target of 4×10^5, one microscan, and a 50 ms maximum injection time. Two hundred ng of HeLa digest was loaded on the column.

Experimental data previously obtained for a 500 ng HeLa injection in both MS1-only and MS/MS-based data-dependent acquisition (DDA) modes using a Q Exactive HF mass spectrometer (Thermo Fisher Scientific, Bremen, Germany) were taken from the PRIDE repository[39] (repository identifier PXD015669) and used for testing upgraded DirectMS1 software. Experimental data for a 1 μg HeLa injection in DDA mode using gradients in a range of 7−60 min were published elsewhere[40] (repository identifier PXD006932). Supplementary Table S1 provides a list and the annotations for all experimental data used in this work.

Data Analysis

Raw files were converted into mzML format using *msConvert* from ProteoWizard[41] (v. 3.0.20066). Peptide features were detected using either Dinosaur[42] (v. 1.2.0) or Biosaur[43] (v. 1.0.3) software for data obtained without and with FAIMS, respectively. Proteins were identified by using ms1searchpy[28] (v. 2.0.3), which is the search engine of DirectMS1 (freely available at https://github.com/markmipt/ms1searchpy under Apache 2.0 license). Parameters for the search were as follows: a minimum of three scans for the detected peptide isotopic cluster, a minimum of one visible C13 isotope, charges from

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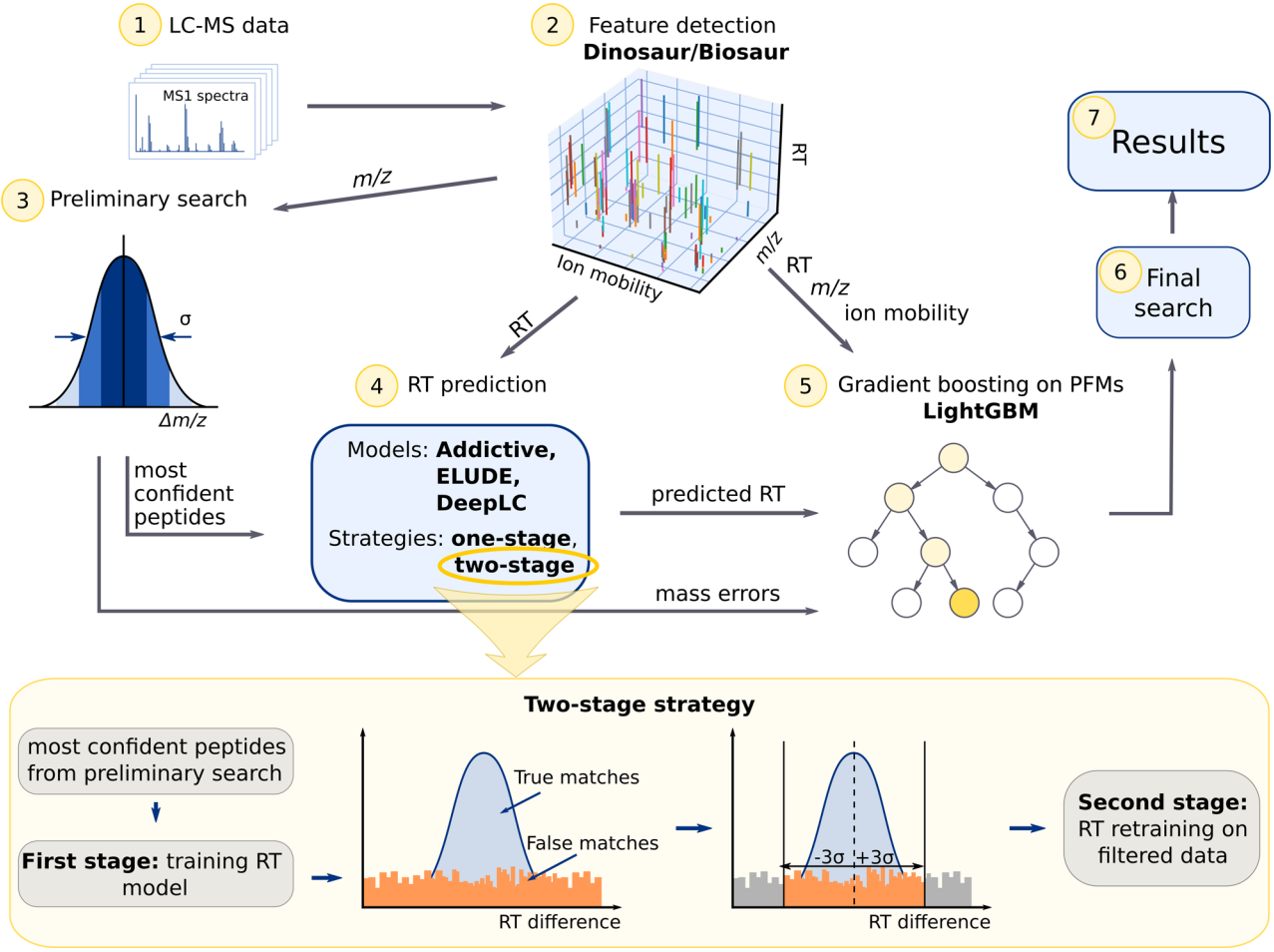


Figure 1. General scheme of the workflow for the DirectMS1 method. The upgraded version of the method's search engine includes the Biosaur algorithm for peptide feature detection in MS1 spectra with the capability of ion mobility data processing, the two-stage RT prediction based on the DeepLC model, and the gradient boosting machine-learning algorithm LightGBM.

1+ to 6+, no missed cleavage sites, and 8 ppm initial mass accuracy. All searches were performed against the Swiss-Prot human concatenated database containing 20 247 protein sequences and its decoys unless otherwise stated. Results were filtered to the 1% protein level false discovery rate (FDR) using the target-decoy approach (TDA)[44] with its "picked" realization[45] and "+1" correction.[46] In the common TDA, the FDR filtering was applied to all target and decoy proteins from the database. In the "picked" realization of the TDA, only the proteins with the highest scores from the pairs of targets and its decoy sequence were left for the FDR filtering.

Data Availability

The data sets generated and analyzed during the current study have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository with the data set identifier PXD023977.

RESULTS AND DISCUSSION

Workflow

Figure 1 shows details of the DirectMS1 method. The method starts with acquiring high-resolution peptide ion mass spectra. A mass spectrometer operates in MS1-only mode and simply collects spectra for eluting peptides at the speed determined by the AGC and mass resolution settings. The total number of MS1 spectra acquired per LC-MS run in the current study was in the range from 1000 to 1500.

The second step addresses peptide feature detection using either Dinosaur or Biosaur utilities. The latter was recently developed in-house and allows ion mobility data, such as FAIMS, to be handled. The third step is a preliminary search performed using only the information about the measured peptide m/z values. High-confidence peptide feature matches (PFMs) (the ones belonging to the proteins identified at 1% FDR in the preliminary search and having the smallest mass error) are selected for the RT prediction model calibration in the next step. By default, the algorithm takes 2500 PFMs with the highest confidence for the calibration, although one should understand that some of these matches are false. (See the discussion that follows.)

Training of peptide RT prediction models integrated into the search workflow is one of the crucial steps in DirectMS1. Originally two RT predictors were available. The first one relied on the so-called "additive" RT prediction model, in which the peptide RT is the sum of the individual retention coefficients of residues constituting the peptide sequence.[47] These coefficients can be either predefined for the type of chromatography used or calculated on-the-fly for specific data and separation conditions. The second model was based on the RT prediction software ELUDE[31] and allowed a significantly more accurate prediction of peptide RTs compared with the

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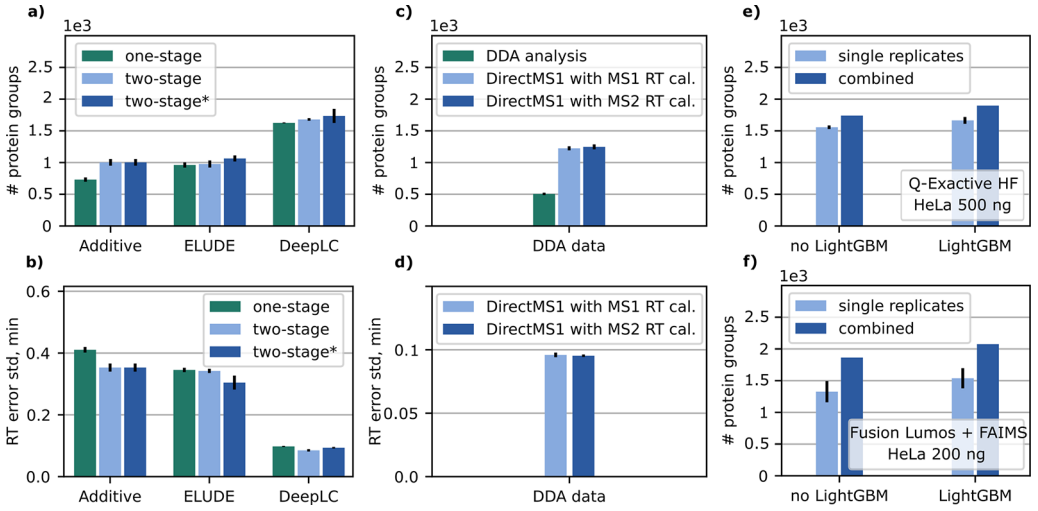


Figure 2. Left column: Analysis of HeLa cell line runs using DirectMS1: (a) Number of protein groups identified at 1% FDR for different RT training strategies. "One-stage" denotes standard RT training, "two-stage" denotes RT training using a two-stage strategy with the same prediction model for both stages, and "two-stage*" denotes RT training using additive and DeepLC models in the first and second stages of training, respectively. (b) Retention time prediction accuracy determined as a standard deviation in the distribution of peptide matches of identified proteins by the difference between experimental and predicted RTs for different RT training strategies. Middle column: Performance of DirectMS1 with different initial training sets for RT prediction: (c) Identified protein groups at 1% FDR. (d) Standard deviation of the difference between the experimental and predicted retention times using DeepLC for reliable peptide matches. The results are shown for the DirectMS1 analysis of the HeLa cell line using DDA data with excluded MS/MS spectra. DDA MS/MS analysis results are shown in green. Right column: Number of identified protein groups at 1% FDR for DirectMS1 with the gradient boosting model, LightGBM, turned on/off: (e) Q Exactive HF and (f) Fusion Lumos with FAIMS setup. Results are shown as an average of three replicates and for all replicates combined into single identification results.

additive model, although at the expense of longer computational time. In this work, we improved the DirectMS1 search module by integrating it with the recently introduced deep-learning RT prediction model, DeepLC. It performs reportedly as well as, or better than, the other state-of-the-art RT prediction models employing machine-learning algorithms. Indeed, we found that DeepLC significantly improved the accuracy of the peptide RT prediction in DirectMS1 compared with the two previously mentioned models that resulted in a dramatic increase in the number of identified proteins, as shown in Figure 2a,b. Furthermore, a two-stage calibration strategy was implemented (Figure 1), which allowed the RT prediction error to be decreased and the number of identifications to be further increased, as detailed in the Retention Time Prediction Models section.

In the fifth step of the workflow, a gradient boosting algorithm is applied to estimate the PFMs quality. The final search uses PFM quality information for protein score calculations and prepares a protein identification list at the specified FDR level, typically 1%. (See the discussion that follows.)

Retention Time Prediction Models

The RT prediction procedure in DirectMS1 has an important distinction compared with that typically used in conventional MS/MS-based searches. The latter usually uses a list of high-confidence peptide-spectrum matches (PSMs) (typically filtered to the 1% FDR level) for RT model training, retraining, or calibration. On the contrary, identifications in DirectMS1 have a high rate of false-positives among PFMs, and it is not possible to obtain a subset of high-confidence peptides of the size reasonable for RT training. DirectMS1 begins with a preliminary search, as shown in Figure 1, as the second step of the workflow using only m/z values of detected PFMs. Then, it forms a preliminary list of identified proteins at 1% FDR. Each of these proteins contains a set of PFMs, with some of them being wrong, and these PFMs have different levels of deviation of the experimentally determined m/z and from the theoretically calculated m/z. PFMs with the lowest mass deviations, up to 2500 PFMs in case of the human cell proteome analyses, are further used for training the DeepLC model. Also note that DirectMS1 uses RT prediction models in different ways. Whereas the additive model is trained from scratch with no use of pretraining, DeepLC is pretrained and then recalibrated for the current data. Importantly, the results of these recalibrations can be affected by the presence of false-positives matches in the training set of PFMs. To alleviate this problem, DirectMS1 employs two-stage RT calibration, as shown in Figure 1, bottom panel. First, the model is trained using all 2500 peptides from the preliminary training set. Then, the software calculates the differences between the predicted and experimental RTs and fits this difference array with the sum of Gaussian and uniform distributions. Here we assume that true peptide matches should follow the Gaussian distribution, whereas the false matches follow the uniform distribution. Using the standard deviation obtained for the distribution of all PFMs by differences between experimental and predicted RTs, peptide matches with RT differences exceeding three standard deviations are excluded from the training set. Then, the model is subjected to final training using the updated PFM set. Figure 2a,b illustrates how different RT prediction models and one- or two-stage training procedures affect the performance of the DirectMS1 method. The results show that the additive model benefits the most from the two-stage RT training process, although it still underperforms compared with machine-learning-based approaches, which are probably less influenced by the presence of false matches in the training set. However, the two-stage RT training where the first stage is performed using the additive model followed by training DeepLC on the set of RT-filtered PFMs, as previously

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described, provides the maximal number of identified proteins. This two-stage training involving two different RT prediction models is denoted as *two-stage** in Figure 2 and is currently used by default in DirectMS1's search engine. All further results presented have been obtained using the *two-stage** training.

Peptide Set for Retention Time Prediction Models

DirectMS1 identifies proteins without the use of any MS/MS spectra. However, one can expect that having MS/MS-based identifications for RT model training may potentially benefit the DirectMS1 analysis. This training could be done in two ways: (i) All data are acquired in DDA mode, and MS1 spectra are then processed by DirectMS1, whereas MS/MS identifications are used for the RT model calibration, and (ii) a separate single DDA run is performed, which is used for RT model training for several subsequent MS1-only experiments. Both approaches have some drawbacks. A separate DDA run requires additional experimental time and samples and additional complications with the need for RT alignment.[48] On the contrary, analyzing data using the standard DDA mode reduces the number of available MS1 spectra, reducing the DirectMS1 performance, or requires longer gradient times for the whole proteome characterization, undermining the concept of high-throughput proteomics. Also, both of the previously mentioned approaches for RT model calibration require a mass spectrometer with MS/MS capabilities, whereas the DirectMS1 method can be used with simpler and more robust MS1-only instrumentation. To further elaborate on this issue, we calculated the number of identified protein groups at a 1% FDR and the RT prediction error obtained using the DeepLC model for different approaches, as shown in Figure 2c,d. DirectMS1 applied to DDA data with excluded MS/MS shows that RT calibration using peptides identified in MS1 analysis is as good as the one based on the peptides identified from MS/MS spectra. Thus there is no need for dedicated DDA runs for RT calibration.

Machine-Learning Analysis of Peptide Feature Matches

Distributions of the mass and RT errors of PFMs are shown in Figure 3a,b for target and decoy matches. Expectedly, the mass error does not provide a strong differentiation between target and decoy PFMs, as shown in Figure 3a. On the contrary, the RT difference between experimental and predicted values potentially presents a strong discrimination metric because peptide RTs are sequence-specific (Figure 3b). Note that there can be some other peptide properties discriminating targets and decoys. For example, the RT prediction and mass measurement errors can depend on the peptide lengths and signal intensities, respectively. Gradient boosting algorithms should potentially account for these nonlinear effects and use them to increase the discriminative power of the search engine. In DirectMS1's search engine *ms1searchpy*, we employed the LightGBM gradient boosting framework for estimating the PFM quality score. A list of target-decoy discriminative properties used for LightGBM in the current version of *ms1searchpy* is shown in Supplementary Table S2. Hyperparameters for LightGBM are tuned in DirectMS1 on-the-fly during data processing. It is implemented using the *Random Search Cross-Validation* method with 25 random hyperparameters testing and three-fold group-based cross-validation. Unique peptide sequences are chosen as groups to prevent overfitting by the presence of the same sequences in both training and test cross-validation sets. The optimal hyperparameters that provide the maximal average number of PFMs at 25% FDR in the test groups are chosen for the final LightGBM model training. Calculated values from gradient boosting are further used in the iteration procedure for the protein score calculation. First, protein binomial scores are calculated using 10% of the PFMs with the best LightGBM scores. In the next iteration, 20% of the PFMs are selected, and scores are calculated again. This continues with 10% steps up to 100%; finally, 10 intermediate protein scores become available for each protein. These scores are averaged and used as the final protein score for target-decoy filtering and for generating the list with protein identification results.

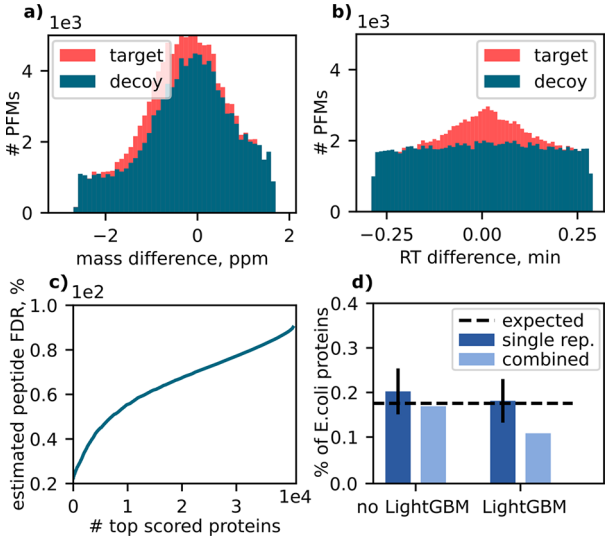


Figure 3. DirectMS1 analysis for 5 min of LC-MS data of 200 ng of HeLa lysate using Lumos with the FAIMS interface. (a) Distributions of target and decoy PFMs by mass difference between theoretical and experimental values. (b) Distributions of target and decoy PFMs by retention time difference between experimental and predicted values. (c) Estimated peptide-level FDR for a cumulative set of top scored proteins (including decoys). (d) Percentage of *E. coli* proteins among all HeLa proteins identified at 1% FDR. Here the dark blue color represents the results averaged over three technical replicates, and the light blue color represents the results for a combination of three technical replicates before the protein score calculation. The dashed black line is the expected fraction of *E. coli* proteins in the results (in percentage).

A comparison of the DirectMS1 performance with and without the LightGBM framework is shown in Figure 2e. LightGBM adds 7−10% of protein identifications. Combining three replicates into the single output further increases the number of protein identifications at 1% FDR to nearly 1900 for this particular data set.

Field Asymmetric Ion Mobility Spectrometry Data

FAIMS is an atmospheric pressure ion mobility technique that separates gas-phase ions by the differences in their mobility in high and low electric fields. FAIMS separation is orthogonal to both LC and MS and provides additional online fractionation to further improve the dynamic range of the analysis of complex samples.[37] Having this third separation dimension in addition to masses and RTs can be beneficial for the DirectMS1 method despite the decrease in the average number of MS1 scans per peptide feature when multiple compensation voltages are used. Although the prediction models for FAIMS data are yet to be developed, adding this

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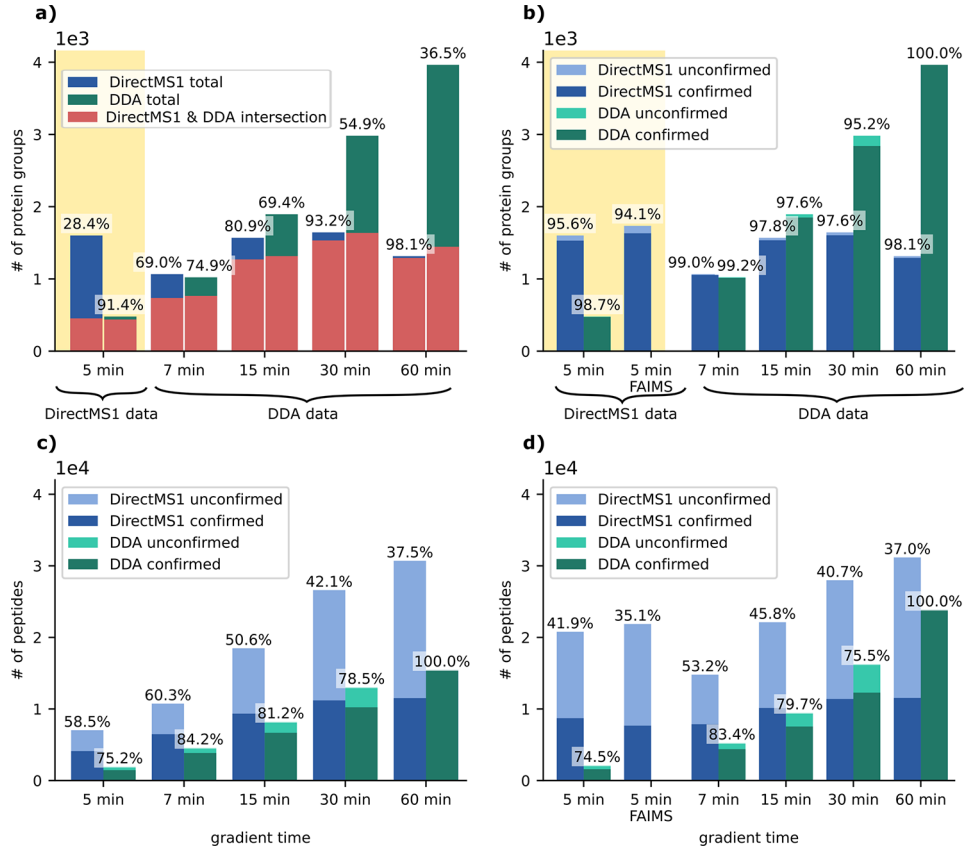


Figure 4. Analysis of HeLa cell line runs using DirectMS1 and IdentiPy[49] + Scavager[50] for different LC gradients. (a) Number of identified protein groups and the intersection between protein groups identified by DirectMS1 and DDA analysis within the same gradient. Percentage values are the ratio of the number of common protein groups (red color) to the number of proteins groups identified by either DirectMS1 (blue) or DDA (green). (b) Number of identified protein groups and the intersection with protein groups identified in a 60 min DDA analysis. Percentages correspond to a fraction of the identified protein groups confirmed by a 60 min DDA analysis. (c) Number of matched peptides and intersection with peptides identified in a 60 min DDA analysis. DirectMS1 and DDA results contain only the peptides belonging to the intersection of proteins identified in both DirectMS1 and DDA results within the same LC gradient at 1% FDR. Percentages correspond to a fraction of matched peptides confirmed by a 60 min DDA analysis. (d) Number of matched peptides and the intersection with peptides identified in a 60 min DDA analysis. DirectMS1 results contain only the peptides belonging to the list of identified proteins at 1% FDR. Percentages correspond to a fraction of matched peptides confirmed by a 60 min DDA analysis. A single technical replicate was used for every LC condition. DirectMS1 results for a 5 min run were obtained using MS1-only data, whereas the rest of the results were obtained using DDA data. The 5 min FAIMS LC data is missing in panels a and d because FAIMS experiments were not performed in DDA mode. The 5 min LC data are from two different runs: the LC-MS1 run for DirectMS1 analysis and the DDA run for DDA analysis, whereas the rest of the LC gradients are from all DDA runs.

dimension into the DirectMS1 workflow resulted in a meaningful increase in the number of identified proteins. As shown in Figure 2f, we were able to identify 2100 proteins at 1% FDR in 5 min of HeLa proteome analysis after combining three technical replicates.

Effect of the Sampling Rate for Feature Detection

The efficiency of the DirectMS1 approach was evaluated for different minimum numbers of MS1 scans in which the peptide features were observed. A single replicate of 5 min of LC-MS1 analysis of HeLa with and without FAIMS was used for the evaluation. Without FAIMS, DirectMS1 reported 1659, 1664, and 1687 protein groups at 1% FDR for a minimum of three, four, and five scans, respectively. For data acquired using FAIMS, we found 1775, 1414, and 961 protein groups for a minimum of three, four, and five scans, respectively. The FAIMS results obtained in this work indicate that lowering a threshold for a minimum number of scans per detected feature is important for 5 min FAIMS-based analyses. Indeed, because FAIMS in our work was operating with three sequentially changed compensation voltages for ion separations, setting a high minimum number of scans per peptide feature will be applied to all three ion mobility separation windows, thus cutting off a large number of legitimate features from being detected. This is further illustrated in Supplemental Figure S1, which shows the distributions of detected peptide features by the number of scans. Hence, to preserve the feature detection efficiency when using FAIMS, the threshold for the minimum number of scans per feature should be lowered compared with that found optimal in the results obtained without FAIMS. For the consistency, the comparison of the results obtained with and without FAIMS was performed for a minimum of three scans as a threshold in both cases. Further exploring the issue with the sampling rate for peptide feature detection is needed to increase the efficiency of the method. We assume that the effect from increasing the minimum number of scans will be more profound for analyses performed using longer LC gradients or fast high-resolution mass analyzers such time-of-flight.

Validation. The FDR at the peptide level can be estimated by the following formula

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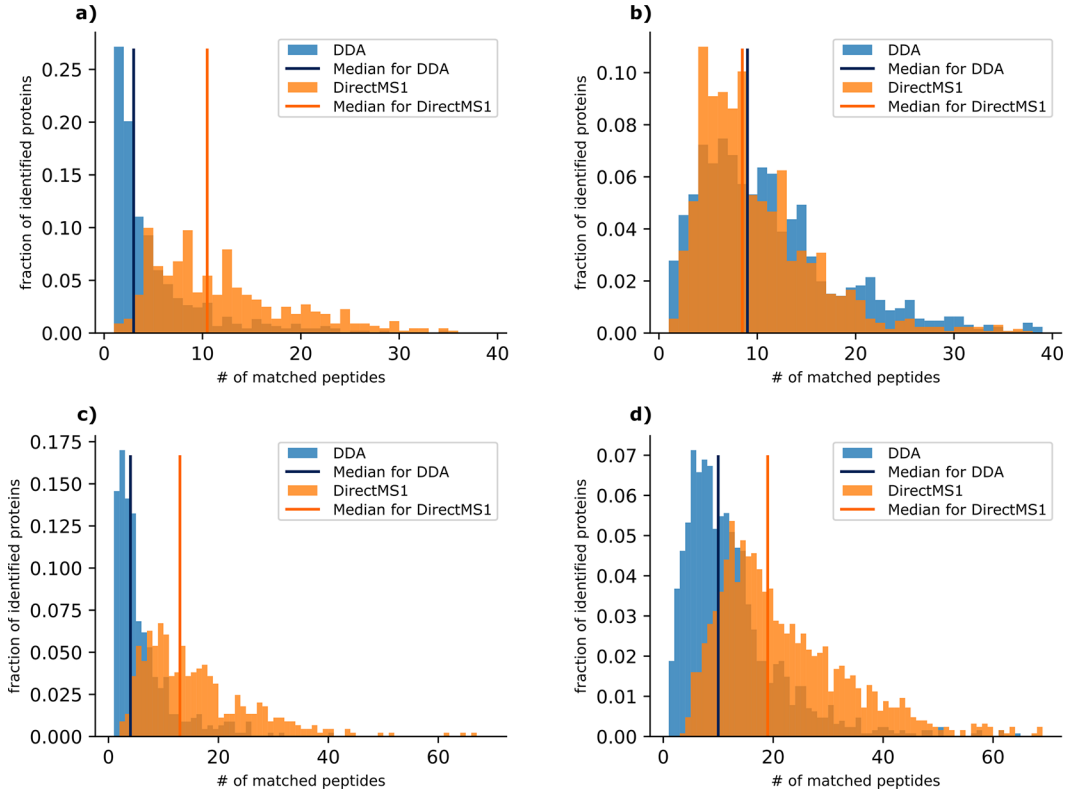


Figure 5. Distributions of identified protein groups by the number of matched peptides for 5 min (a,c) and 60 min (b,d) LC gradient data sets. (a,b) Peptides identified at 1% FDR in DDA and the corrected number of peptides matching the identified proteins in DirectMS1. (c,d) All peptides identified in DDA without FDR filtering and the total number of peptides matching the identified proteins in DirectMS1 without correction. For a fair comparison, the results are shown only for protein groups that are common for DDA and DirectMS1 (454 and 1290 proteins identified in 5 and 60 min of data, respectively).

$$\mathrm{FDR}_{\mathrm{peptide}} = p \times \frac{\text{number of theoretical peptides}}{\text{number of matched peptides}} \quad (1)$$

where

$$p = \frac{\text{number of matched decoy peptides}}{\text{number of theoretical decoy peptides}} \quad (2)$$

The FDR analysis at the peptide level is shown in Figure 3c. Strikingly enough, the average FDR for all peptide feature matches is close to 90%. However, the average peptide-level FDR values vary from 23 to 32% for the top-100 scored proteins and for all proteins identified at 1% FDR, respectively.

The FDR at the protein level is estimated as follows

$$\mathrm{FDR}_{\mathrm{protein}} = \frac{\text{number of picked decoy proteins} + 1}{\text{number of picked target proteins}} \quad (3)$$

Whereas the protein FDR level is estimated using TDA, the other interesting question is how many proteins from the target database not present in the sample can be found in the results by DirectMS1's search engine? To perform this control analysis, HeLa data were searched against the combined human and *E. coli* Swiss-Prot databases. The percentage of identified *E. coli* proteins in DirectMS1 analysis is shown in Figure 3d. The number of identified *E. coli* proteins was in the range of two to three sequences out of 1124 to 1173 proteins identified at 1% FDR for the analyzed sample. (See Supplementary Table S3.) The expected percentage of *E. coli* proteins was calculated using the following equation

$$\text{expected percentage of } \textit{E. coli} \text{ proteins} = \mathrm{FDR} \times \frac{\text{number of } \textit{E. coli} \text{ proteins in database}}{\text{total number of proteins in database}} \quad (4)$$

that corresponds to 0.18% of *E. coli* proteins at 1% FDR in the identification results. (The *E. coli* and human databases contain 8862 and 40 386 sequences including decoys, respectively.) These expectations are in good agreement with the experimental results that further confirm the validity of the protein identification filtering algorithm implemented in DirectMS1.

To further explore this important issue, we first looked at the intersection between the protein groups identified by DirectMS1 and those identified by standard MS/MS-based DDA analyses shown in Figure 4. First, we found that in the case of a 5 min gradient analysis, close to 90% of the proteins identified by DDA are the same as those identified using the DirectMS1 method with MS1-only data acquisition, as shown in Figure 4a. Note that the efficiency of the standard DDA approach in 5 min analyses did not exceed 500 proteins, and most of these proteins were identified by one or two peptides.

It would be more interesting to see how many protein identifications in DirectMS1 can be validated using MS/MS-based analyses. This validation was performed by counting the number of proteins identified at 1% FDR by DirectMS1 that are confirmed by long-gradient DDA analysis. We used proteins identified in the 60 min LC gradient MS/MS-based analysis of HeLa standard digest as the "ground truth" and compared them with the protein identifications in DDA data obtained using 7, 15, 30, and 60 min LC gradients. (See the

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data details in the Experimental Section.) The results of the comparison are shown in Figure 4b. For each gradient length, we identified proteins either using only MS1 spectra and the DirectMS1 search engine or performing standard MS/MS-based searching. Additionally, the 5 min LC-MS1-only results obtained by the DirectMS1 method with and without FAIMS were added to the comparison. Figure 4b shows that 95% of the proteins identified in the 5 min DirectMS1 analysis were also identified in the 60 min DDA MS/MS-based experiments. As for the 7, 15, 30, and 60 min LC gradients, an intersection between MS1-only searches and 60 min DDA results was within 97.6−99.0%. For comparison, intersections between MS/MS-based identifications for different LC gradients with 60 min DDA proteins were in the range of 95.2−99.2%. These results validate the FDR estimation performed by the DirectMS1 method and its search engine at the protein level. A similar analysis was performed for peptides. The peptides identified in the 60 min DDA analysis were used as the "ground" truth peptides. As Figure 4d shows, only 35−50% of "DirectMS1" peptides were confirmed by the "ground" truth peptides. Importantly, the rest of the "DirectMS1" peptides are not necessarily the false matches. A significant portion of them are true peptides corresponding to valid protein identifications. Indeed, DDA analysis identifies only peptides selected for fragmentation. DirectMS1 takes into account all peptides detected in MS1 spectra. Thus the rest of the peptides shown in the light blue color in Figure 4d constitute false matches and true peptides unique for the DirectMS1 analysis.

Finally, to estimate the protein sequence coverage in the DirectMS1 method and compare it with MS/MS-based DDA, only peptides of the proteins identified by both DirectMS1 and DDA for the same LC gradient data were considered. As Figure 4c shows, DirectMS1 reports more peptide matches than DDA for all LC gradients, even after correction on an estimated peptide-level FDR of ∼32%.

Comparison of Protein Sequence Coverage in DirectMS1 and DDA Methods

The protein sequence coverage for DDA and DirectMS1 methods is shown in Figure 5 as the distributions of peptides identified per protein. The median values of the number of peptides identified per a protein at 1% FDR using DDA were 3 and 9 for 5 and 60 min of data, respectively. The median values of the corrected number of peptides matched per a protein in DirectMS1 were 10.5 and 8.5 for 5 and 60 min of data, respectively. The latter correction was performed by subtracting the estimated number of false peptides matches from the total number of matched peptides. The estimation of the number of false peptides was based on the calculation of the peptide FDR previously described (eq 1). For example, if a peptide-level FDR is 20% and a protein has 10 peptide matches, then 8 peptide matches is used for the above comparison. To further extend the comparison, we calculated the peptide distribution for the identified proteins without FDR filtering in both methods (Figure 5c,d). These results show that even without filtering, DDA does not provide comparable sequence coverage for identified proteins, thus further alleviating the question regarding the reason for the higher protein sequence coverage in the DirectMS1 method due to the higher FDR of peptide matches. Expectedly, the longer the gradient time used for the analysis, the closer the sequence coverage in both methods.

CONCLUSIONS

DirectMS1 for high-throughput MS/MS-free quantitative proteome analysis was upgraded with DeepLC, the novel machine-learning-based utility for peptide retention time prediction and the recently introduced gradient boosting framework LightGBM for scoring peptide feature matches in MS1 spectra. Additionally, two-stage and two-model peptide retention time prediction was included into the method's workflow. The method was then evaluated using a high-resolution mass spectrometer with a FAIMS interface. These additions resulted in a two-fold increase in the efficiency of the DirectMS1 method, which is now capable of identifying as many as 2000 proteins at 1% FDR in a 5 min MS1-only analysis of cellular proteomes. Further improvements in the depth of proteome coverage using this method are expected with improvements in ion mobility data processing and prediction.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jproteome.0c00863.

Supplementary Table S1. Raw files description with experimental parameters (XLSX)

Supplementary Table S2. List of target-decoy discriminative properties used for LightGBM (XLSX)

Supplementary Table S3. A number of identified *E. coli* and human proteins for HeLa data searched against combined human and *E. coli* Swiss-Prot databases (XLSX)

Concept behind protein scoring and FDR estimation in the DirectMS1 method (PDF)

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Notes

The authors declare no competing financial interest.
The data sets generated and analyzed during the current study have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository with the data set identifier PXD023977.

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